

ARTICLE

The developmental consequences of early exposure to climate change-related risks

Jorge Cuartas^{1,2}  | Dana C. McCoy²  | Isabella Torres² | Lindsey Burghardt³ | Jack P. Shonkoff^{2,3,4,5,6} | Hirokazu Yoshikawa⁷

¹Centro de Estudios sobre Seguridad y Drogas (CESED), Universidad de los Andes, Bogota, Colombia

²Harvard Graduate School of Education, Cambridge, Massachusetts, USA

³The Center on the Developing Child at Harvard University, Cambridge, Massachusetts, USA

⁴Harvard T.H. Chan School of Public Health, Boston, Massachusetts, USA

⁵Harvard Medical School, Boston, Massachusetts, USA

⁶Boston Children's Hospital, Boston, Massachusetts, USA

⁷NYU Global TIES for Children, New York, New York, USA

Correspondence

Jorge Cuartas, Harvard Graduate School of Education, 13 Appian Way, Cambridge, MA 02138, USA.
Email: jcuartas@g.harvard.edu

Abstract

The climate crisis encompasses a constellation of risks that threaten human livelihoods, well-being, and survival globally. In this article, we present a new framework based on bioecological and dynamic systems perspectives, and on evidence for conceptualizing how the distinctive dual time frame of both acute (e.g., extreme weather events) and chronic (e.g., ecological degradation) climate change-related risks experienced prenatally and early in life across multiple ecological contexts can threaten human development. We conclude with a call to developmental researchers, practitioners, and policymakers to invest more efforts in understanding and addressing the climate crisis and its developmental consequences to ensure a sustainable future for all.

KEYWORDS

climate change, climate crisis, early childhood development, prenatal development

The climate crisis encompasses a constellation of current and future risks that threaten human livelihoods, well-being, and survival globally. Humankind has increasingly faced devastation, morbidity, mortality, and scarcity of resources, as well as profound changes in standards of living due to ecological degradation, extreme weather events, heat waves, and infectious diseases, among other impacts related to climate change (Romanello et al., 2022). Action and inaction by governments and societies may lead to an increase of 2°C or more in the average global temperature by the end of the century (Pörtner et al., 2022; Sherwood et al., 2020), exacerbating multiple existential risks and deepening current humanitarian crises.

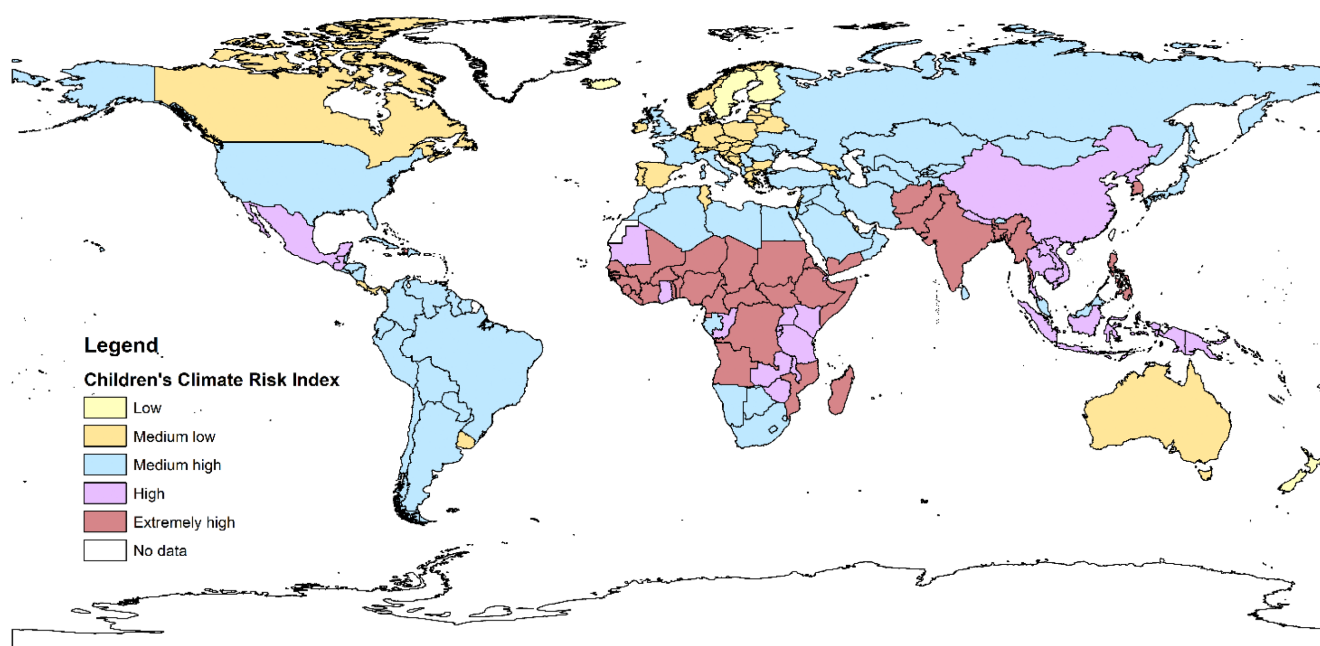
Young children have been described as being affected first and worst by the climate crisis (Rees, 2021). Almost every child on earth is already exposed to climate and environmental hazards, and about 1 billion (or 50% of)

children are currently living in countries categorized as high in climate risk due to these exposures and inadequate access to essential health, education, and social protection services (Rees, 2021; see Figure 1). The climate crisis may be particularly harmful when experienced prenatally and during early childhood because of heightened neurobiological plasticity, which makes the developing brain and other biological systems especially vulnerable to biological and psychosocial risks (Black et al., 2017; Nelson & Gabard-Durnam, 2020; Wachs et al., 2014). Nonetheless, developmental research has focused little attention on the current and potential impacts of the developing climate crisis on young children relative to adolescents and adults (Burke et al., 2018).

In this article, we present a framework based on bioecological and dynamic systems perspectives to conceptualize the theoretical mechanisms through which risks related to the climate crisis that are experienced early in



Climate Risk Index



Population of children younger than five

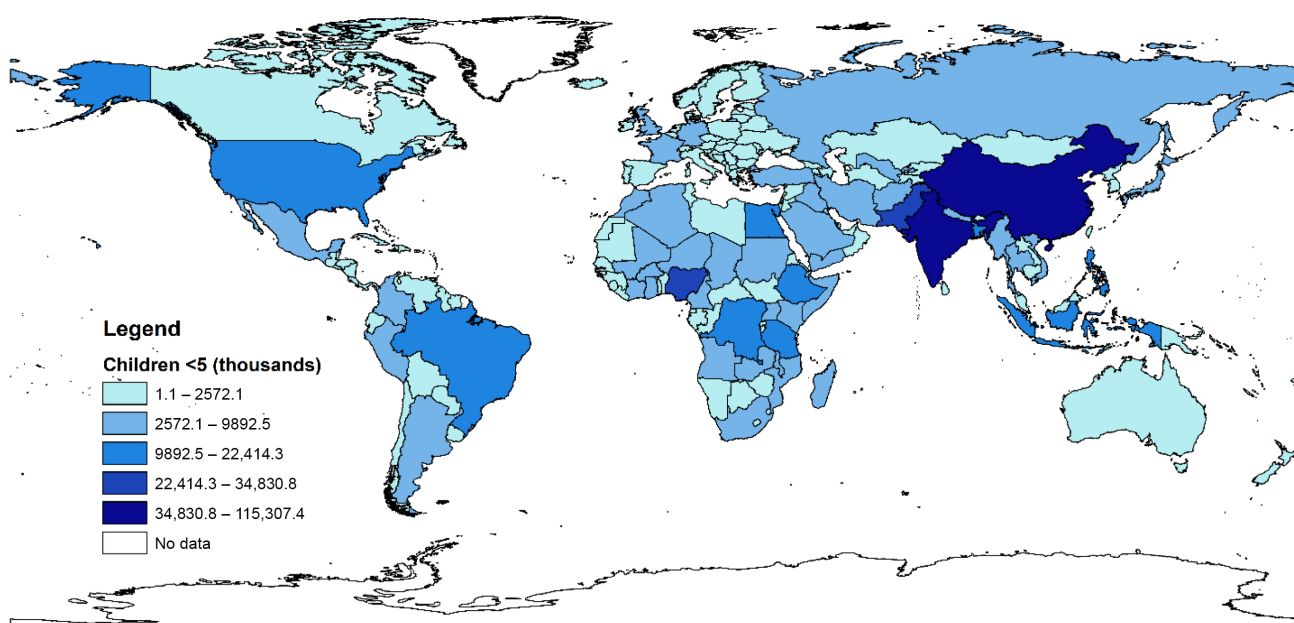


FIGURE 1 Children's climate risk index (top) and population of children younger than 5 (bottom), by country. Data from UNICEF (Rees, 2021) and *World Population Prospects* (United Nations, 2017). The Children's Climate Risk Index is computed by considering children's exposure to environmental shocks (e.g., heat waves) and their susceptibility based on lack of access to essential services (e.g., water).

life (i.e., prenatally and in early childhood, up to age 5) may influence human development across the life course. This framework reflects two key features that differentiate climate change from other macro-level crises studied in developmental science: (1) its inclusion of both acute shocks and protracted risks (i.e., its chronicity), and (2) its influence across multiple ecological contexts that

include communities, families, and individuals (i.e., its pervasiveness). Since some of these risks move at a slow pace and are not readily visible in certain settings, our proposed framework aims to highlight opportunities to better understand and address the systemic roots and proximal factors of climate-related risks for children's development.



A FRAMEWORK FOR UNDERSTANDING CLIMATE CHANGE-RELATED RISKS AND DEVELOPMENTAL OUTCOMES

In developmental research, few studies have addressed the developmental consequences of experiencing risks related to climate change prenatally and during early childhood, the *mechanisms* underlying these associations, and the *moderators* that may explain differential outcomes for affected individuals. Exploring these outcomes, mechanisms, and moderators is critical for building more robust theories about the climate crisis and human development, as well as for identifying potential targets for intervention. We use bioecological and dynamic systems perspectives to conceptualize the mechanisms linking children's early exposure to the climate crisis and their developmental outcomes, as well as the moderators of these processes.

Bioecological and dynamic systems theories conceptualize development and behavior as the probabilistic outcome of dynamic, gradually more complex interactions among biological and environmental processes at multiple levels of the ecology, in many systems, and across time (Yoshikawa & Hsueh, 2001). They include four key principles: First, individuals and their ecologies are mutually interdependent and capable of transforming one another (i.e., reciprocity). Second, human development progresses toward greater organizational complexity over time (e.g., dynamic development of skills that allow for progressively more complex interactions with the physical and social worlds; Sameroff, 2010). Third, systems tend to operate most of the time within a constrained set of parameters (e.g., biological constraints). Finally, changes within systems can occur in both gradual and discontinuous (transformational) ways, depending on the magnitude of change in parameters and whether parameters exceed critical thresholds that lead to the reorganization of the whole system (Thelen & Smith, 1998; Yoshikawa & Hsueh, 2001).

Reflecting these core principles, our framework (see Figure 2) posits that exposure to the climate crisis early in life threatens human development through its far-reaching acute impacts and protracted changes on key mechanisms (e.g., air pollution, destruction of infrastructure, violence, caregiver stress) at all levels of the bioecological model (i.e., environment, societies, communities, families) that may lead to dynamic transformations in systems over time (see Mechanisms sections). Our framework also shows that the climate crisis will likely act as a threat multiplier, leading to worse consequences for communities and families facing pre-existing adversity (see Moderators section). Reflecting the reciprocity principle, the framework also highlights the importance of promoting healthy prenatal and early childhood development to raise generations of children

who can contribute to mitigating the crisis and promoting resilient communities and societies. Next, we highlight several theoretical mechanisms of transmission and moderators, discussing the distinctive influences of acute and protracted risks at the physical environment, society, community, and family levels of the developmental ecology.

Mechanisms at the physical environmental level

The climate crisis is not a single event, but rather a constellation of ecological risks resulting from global rises in temperature (Romanello et al., 2022). These ecological risks can affect prenatal and child development through changes in *physical environments*. For example, acute extreme weather events can lead to immediate morbidity and mortality (Helldén et al., 2021) and reprogram the placental molecular signature when experienced during pregnancy, with downstream consequences for lifelong stress regulation and emotional health, in line with a fetal programming hypothesis (Hoche, 2014; Nomura et al., 2021). Protracted and more permanent changes in vector (i.e., transmitters of disease-causing organisms) patterns, increased pollution, and lower availability and quality of water and fertile land can also heighten children's risk of disease (e.g., asthma), malnutrition, and atypical brain development both prenatally and during the first years of life, undermining the development of foundational skills (Berens & Nelson, 2019; Bové et al., 2019; Helldén et al., 2021; Peterson et al., 2022; Walker et al., 2011). To understand more completely the potential developmental consequences of early-life exposure to climate-related risks in physical environments, we reviewed more than 100 studies on the associations of experiencing pollution, extreme weather events, ecological degradation, and zoonotic diseases prenatally and during early childhood with developmental outcomes (see Supporting Information S1 for the search strategy and Table S1 for the sociodemographic characteristics of the studies). Most of the studies were conducted in high-income countries, with only a few in low- and middle-income countries.

Exposure to climate change-related risks in physical environments prenatally and the first years of life can undermine foundational cognitive, motor, social-emotional, and behavioral skills, and compromise lifelong developmental trajectories. Most published research on this topic has focused on understanding the links between air pollution and developmental outcomes; these studies have found that prenatal and postnatal exposure to outdoor and indoor air pollution is associated with adverse cognitive, motor, social-emotional, and behavioral development, including behavior regulation, inhibitory control, and internalizing problems (e.g., Bansal et al., 2021; Odo et al., 2023). Other studies have

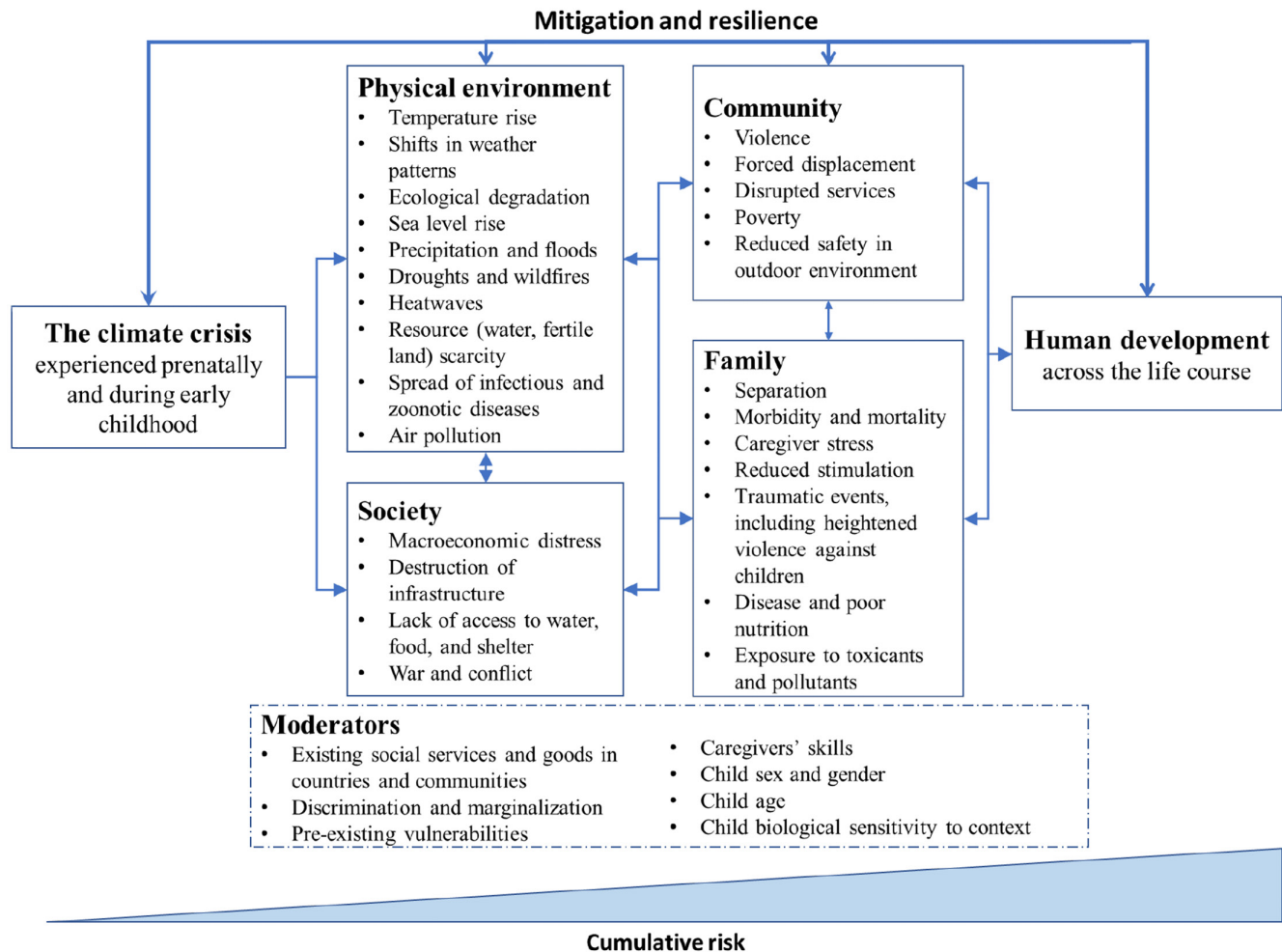


FIGURE 2 A bioecological and dynamic systems perspective on the climate crisis and early childhood development.

identified associations between early exposure to other pollutants (e.g., heavy metals, persistent organic pollutants) and negative cognitive, socioemotional, and motor outcomes for children in early (e.g., Kim et al., 2018; Niu et al., 2019) and middle childhood (Gaspar et al., 2015; van den Dries et al., 2021).

Early exposure to extreme weather events like floods, hurricanes, and extreme temperatures has also been linked to adverse behavioral outcomes (e.g., Simcock et al., 2019), motor development (e.g., Moss et al., 2017), cognitive development (e.g., Guo et al., 2020), and emotion regulation and temperament (Buthmann et al., 2019) in early childhood, as well as reduced prosocial behavior in middle childhood (Duchoslav, 2017) and increased risk for anxiety disorders in adulthood (Maclean et al., 2016). Many studies have also found positive associations between exposure to green spaces and cognitive development and self-regulation (e.g., Almeida et al., 2022; Liao et al., 2019), implying negative consequences of climate-related ecological degradation. Finally, evidence suggests that prenatal exposure to zoonotic diseases, like the Zika virus, is linked to impaired early language development (Valdes et al., 2019).

Mechanisms at the society level

The physical devastation caused by the climate crisis can lead to changes at the *society* level, including acute economic shocks and more protracted macroeconomic distress, inflationary pressures, and worsened inequalities driven by intergenerational poverty, systemic racism, and other deeply embedded structural inequities (Campbell & Spencer, 2021; Ibarrarán et al., 2009). These societal impacts, in turn, are likely to further threaten the capacity of states, communities, and families to provide adequate prenatal care, nutrition, health, safety, and learning opportunities for children, with consequences for developmental outcomes (Duncan et al., 2015; Jensen et al., 2017). Indeed, research has already identified sizable socioeconomic gradients in children's exposure to learning opportunities in the family system (e.g., reading, playing) and outside the home (e.g., attendance in early childhood education programs) and young children's foundational cognitive and social-emotional skills between and within countries (Lu et al., 2020; McCoy et al., 2016). With economic resources at multiple ecological levels being stretched thin over time, these



gaps will likely widen even further as the climate crisis deepens (Romanello et al., 2022).

Acute and cumulative extreme weather events, especially when combined with weakened economies, can also lead to society-level destruction of and reduced investments in key infrastructure, transportation networks, and social services (Vallejo & Mullan, 2017). Crises like the COVID-19 pandemic highlighted the fragility of existing services and the negative consequences of limiting access. For example, prolonged closures of early care and education centers at the start of the pandemic have been predicted to compromise children's early development and lifelong learning and economic trajectories (McCoy et al., 2021). Indeed, disruptions in infrastructure and services due to the climate crisis—which is expected to be a far more chronic source of risk than even the COVID-19 pandemic—may impose considerable challenges for children's development given the known developmental benefits of access to water and sanitation, early childhood education, and parenting programs, among others (Barnett & Nores, 2015; Gao et al., 2021; Jeong et al., 2021).

The climate crisis and downstream acute shocks and resource limitations may also lead to protracted and, eventually, discontinuous changes that exacerbate competition, instability, and the risk for conflict and war, especially in settings marked by pre-existing political and economic fragility and marginalization (Breckner & Sunde, 2019; Koubi, 2019; Mach et al., 2019). In studies using data from China (Lee et al., 2017), Syria (Kelley et al., 2015), Somalia (Maystadt et al., 2014), and cross-national samples (Bollfrass & Shaver, 2015; Couttenier & Soubeyran, 2013; Ghimire & Ferreira, 2016; Salehyan & Hendrix, 2014), researchers have found links (in some studies, small in magnitude) between climate-related risks and the outbreak or intensification of civil war. Conflicts increase children's likelihood of exposure to traumatic events that can trigger the onset of mental health problems (Sánchez-Ariza et al., 2023) and undermine the development of early skills through processes of social learning (Bandura, 1986), social information processing (Dodge, 1993), and neurophysiological mechanisms (McLaughlin et al., 2019).

Mechanisms at the community level

Economic turmoil, scarcity of services, and conflict can also affect *communities* by exacerbating forced displacement. Of the record 100 million people worldwide who were forcibly displaced in 2022, most were relocated due to climate-related disasters (UNHCR, 2022). In addition, most young children are at risk of forced displacement as a result of living in areas that experience higher rates of both acute extreme weather events and slower climate-related impacts like rising sea levels (Pörtner et al., 2022; Rees, 2021). Forced displacement is a major risk factor

for multiple forms of adversity that have consequences for early development, including family separation and neglect (Masten et al., 2019).

Furthermore, increased temperatures and levels of pollution can reduce safety in outdoor environments, potentially impeding children's physical activity and learning opportunities within their communities. For example, in one study, on days with higher temperatures, U.S. preschoolers engaged in less moderate to vigorous physical activity and more sedentary behavior than they did on cooler days (Koepp et al., 2023); such movement behaviors have been associated with young children's motor and cognitive development (Kuzik et al., 2020). In addition, barriers to spending time in outdoor spaces (e.g., high temperatures, pollution) can reduce children's opportunities to interact with other children and adults, which is critical for their social-emotional development (Putra et al., 2020; Zare Sakhvidi et al., 2022).

Mechanisms at the family level

Acute and protracted risks exacerbated by crises in societies and communities can also affect *families*, most notably by increasing caregivers' mental health challenges. In one review and a large empirical study of U.S. residents across a decade, both acute climate-related events and more chronic changes (e.g., growing anxiety due to the existential threat of the crisis) were linked to increased mental health challenges in both adults and children (Obradovich et al., 2018; Palinkas & Wong, 2020). Developmental evidence and theories like the family stress model have long demonstrated the influence of caregivers' mental health on their responsiveness to children's cues—and on young children's development in general—especially when the children face adversity and require particularly sensitive caregiving to help them develop resilience (Cicchetti, 2013; Conger et al., 2000; Wachs et al., 2009). Similarly, heightened mental health challenges during pregnancy also affect both cognitive and social-emotional skills in early childhood (Berthelon et al., 2021).

Finally, intensified, climate change-related, contextual stressors and heightened mental health challenges can lead to individual distress and interpersonal aggression and violence (Hsiang et al., 2013; Miles-Novelo & Anderson, 2019, 2022). These impacts may undermine the stable and supportive caregiving that promotes positive adaptation by buffering young children from the disruptive effects of excessive stress activation in the face of significant threat and uncertainty. Structural characteristics and acute shocks within community contexts affect caregiving behaviors and interactions with children related to violence and engagement in early learning opportunities (Cuartas et al., 2018; Cuellar et al., 2015). These caregiving behaviors and interactions, often conceptualized as the proximal processes that drive

development (Bronfenbrenner & Morris, 2007), significantly predict foundational skills globally (Cuartas et al., 2023; Heilmann et al., 2021; Jeong et al., 2017).

MODERATORS: INDIVIDUAL VARIATION AND INEQUITIES

The climate crisis will not affect all communities and families equally and is much more likely to widen existing opportunity gaps beginning early in life. Next, we briefly discuss some individual and contextual characteristics that can *moderate* (i.e., change the strength or direction of) different parts of the causal chain leading from climate change to impaired human development (see Figure 2, Moderators).

Children in low- and middle-income countries and communities with limited access to basic social services, which are considered to be high in climate risk, are the most likely to suffer given their lack of basic resources to adapt to and cope with acute shocks and protracted risks. Children and families who are already experiencing significant adversity (e.g., extreme poverty, violence, discrimination, racism) and those facing pre-existing developmental difficulties might also be at heightened risk due to cumulative burdens that overload both caregivers' capabilities and children's adaptive capacities (Center on the Developing Child at Harvard University, 2016; McLaughlin et al., 2021; Tessum et al., 2019).

Heterogeneity in children's sensitivity to context, particularly in the face of adversity, will also play a significant role in their susceptibility to the climate crisis. For example, different physical (e.g., exposure to pollutants, insufficient micronutrients) and psychosocial (e.g., neglect, climate-related stress) exposures will have differential consequences depending on children's age and the timing of sensitive periods in the development of specific biological systems (Hensch, 2005; Knudsen, 2004; Nelson & Gabard-Durnam, 2020; Wachs et al., 2014). Similarly, evidence on individual variation and the biological sensitivity to context theory (Pluess, 2015) indicate that different children might be more (or less) biologically susceptible to the potential acute and chronic impacts of climate change on their developmental ecologies.

Finally, access to protective factors that may mitigate the negative impacts of the crisis will also vary—and such factors could be useful as targets of interventions. One key candidate is caregivers' resources and capabilities (Leijten et al., 2019; Sandler et al., 2015). Indeed, caregivers with adequate social-emotional and executive function skills are likely to be more resilient and better prepared to support children in the face of extreme weather events and societal unrest (Center on the Developing Child at Harvard University, 2016; Gartland et al., 2019; Sanson et al., 2019). These caregivers' resources and capabilities largely depend on systemic factors that may be affected by the climate crisis, like the availability of social support, high-paying

jobs, affordable housing and child care, and access to food and clean water.

LOOKING AHEAD

The most frequently cited, long-term solution to the climate crisis is to decarbonize economies, particularly in the high-income countries that have historically contributed most to the crisis (Rees, 2021; Romanello et al., 2022). Yet there is also an urgent need to help families and children adapt to the impacts of climate change, especially in low- and middle-income countries that are less resilient to climate change. Indeed, the climate crisis is global in scope and therefore requires solutions that are also global. Developmental researchers can contribute to preparation and adaptation policies and strategies aimed at supporting prenatal and early postnatal development in the face of climate-related threats, but the most compelling need is to educate policymakers and the general public about the urgent need to confront this growing crisis proactively.

The chronicity and pervasiveness of the climate crisis reflect an accumulation of risk across both time and space that is unprecedented in the developmental literature. Therefore, beyond calculating the number of children at risk, developmentalists must build on our proposed framework, and produce and disseminate a clear causal story that explains how exposure to climate-related risks gets into the body and affects the development of the whole child. We also urgently need more research, particularly from less climate-resilient countries, to inform targets of science-informed interventions aimed at promoting resilience and adaptation. Questions that developmental scientists could explore include: How might macro-level strategies to address the climate crisis (e.g., decarbonization efforts) affect children's risks and development? To what extent are more localized and evidence-based mitigation efforts focused on promoting healthy development (e.g., early childhood education, home visiting, cash transfers, increased access to green spaces) sufficient for protecting children from climate-related threats? Which novel, alternative, or complementary approaches may help further alleviate risk?

Regarding policy and practice, the child development community can participate more actively in science-informed advocacy around policies to prevent and mitigate climate change and can support the design and scale-up of interventions that maximize preparedness. For example, developmental scientists can contribute to developing policies and programs that help caregivers and children cope with climate-related disasters by promoting core life skills (e.g., self-regulation, executive function, and attention; Jeong et al., 2021; Leijten et al., 2019; Sanson et al., 2019). Similarly, developmental scientists can support the design of systemic interventions aimed at supporting communities, caregivers, and children to promote climate resilience.

Finally, the dynamic systems and ecological perspectives we have discussed indicate that current and future generations of children will be key actors in mitigating the climate crisis and transforming systems of human development for the better (see Figure 2, Mitigation and Resilience). Developmental scientists could play a critical role in informing and supporting efforts to educate children early on about climate change to raise generations of leaders on this topic (see, e.g., Earth Warriors, earthwarriorsglobal.com). Promoting and supporting children's agency and activism, combined with science-informed interventions, may pay a "triple dividend" (Patton et al., 2016, p. 2424) for children today, in their future adult lives, and for future generations.

In conclusion, the climate crisis poses an unprecedented, enormous threat to the well-being (and indeed the survival) of current and future generations due to its dual time frame by leading to both acute shocks and gradual increases in all risks in multiple developmental contexts (i.e., chronicity and pervasiveness). Developmental scientists can play an important role in explaining the adverse developmental consequences of the climate crisis and in supporting the design and implementation of evidence-informed policies and strategies to support prenatal and early healthy development in the face of this looming threat. Supporting current and future generations of children is critical to addressing long-lasting inequities and human rights issues and to raising generations of leaders to mitigate the climate crisis and ensure a sustainable future for all.

ACKNOWLEDGMENTS

We thank Anna Dorste for her support of the literature review, Cristina Ricarte and Catalina Rey-Guerra for their comments and help, and the editor and two reviewers for their feedback.

FUNDING INFORMATION

This research received no specific grant from any funding agency in the public, commercial, or non-for-profit sectors.

ORCID

Jorge Cuartas  <https://orcid.org/0000-0002-0451-3298>

Dana C. McCoy  <https://orcid.org/0000-0002-0896-1813>

REFERENCES

- Almeida, D. Q., Barros, H., & Ribeiro, A. I. (2022). Residential and school green and blue spaces and intelligence in children: The generation XXI birth cohort. *Science of the Total Environment*, 813, 151859. <https://doi.org/10.1016/j.scitotenv.2021.151859>
- Bandura, A. (1986). *Social foundations of thought and action: A social cognitive theory*. Prentice Hall.
- Bansal, E., Hsu, H. H., de Water, E., Martínez-Medina, S., Schnaas, L., Just, A. C., Horton, M., Bellinger, D. C., Téllez-Rojo, M. M., & Wright, R. O. (2021). Prenatal PM2.5 exposure in the second and third trimesters predicts neurocognitive performance at age 9–10 years: A cohort study of Mexico City children. *Environmental Research*, 202(November), 111651. <https://doi.org/10.1016/j.envres.2021.111651>
- Barnett, W. S., & Nores, M. (2015). Investment and productivity arguments for ECCE. In P. T. M. Marope & Y. Kaga (Eds.), *Investing against evidence: The global state of early childhood care and education*, 73–88. United Nations Educational, Science and Cultural Organization.
- Berens, A. E., & Nelson, C. A. (2019). Neurobiology of fetal and infant development: Implications for infant mental health. In C. H. Zeanah (Ed.), *Handbook of infant mental health*, 41–62. Guilford Press.
- Berthelon, M., Kruger, D., & Sanchez, R. (2021). Maternal stress during pregnancy and early childhood development. *Economics and Human Biology*, 43(December), 101047. <https://doi.org/10.1016/j.ehb.2021.101047>
- Black, M., Walker, S. P., Fernald, L. C. H., Andersen, C. T., DiGirolamo, A. M., Lu, C., McCoy, D. C., Fink, G., Shawar, Y. R., Shiffman, J., Devercelli, A. E., Wodon, Q. T., Vargas-Barón, E., & Grantham-McGregor, S. (2017). Early childhood development coming of age: Science through the life course. *The Lancet*, 389(10064), 77–90. [https://doi.org/10.1016/S0140-6736\(16\)31389-7](https://doi.org/10.1016/S0140-6736(16)31389-7)
- Bollfrass, A., & Shaver, A. (2015). The effects of temperature on political violence: Global evidence at the subnational level. *PLoS One*, 10(5), e0123505. <https://doi.org/10.1371/journal.pone.0123505>
- Bové, H., Bongaerts, E., Slenders, E., Bijmens, E. M., Saenen, N. D., Gyselaers, W., Van Eyken, P., Plusquin, M., Roeffaers, M. B. J., Ameloot, M., & Nawrot, T. S. (2019). Ambient black carbon particles reach the fetal side of human placenta. *Nature Communications*, 10(1), 3866. <https://doi.org/10.1038/s41467-019-11654-3>
- Breckner, M., & Sunde, U. (2019). Temperature extremes, global warming, and armed conflict: New insights from high resolution data. *World Development*, 123, 104624. https://econpapers.repec.org/article/eeewdevel/v_3a123_3ay_3a2019_3ai_3ac_3a24.htm
- Bronfenbrenner, U., & Morris, P. (2007). The bioecological model of human development. In W. Damon & R. M. Lerner (Eds.), *Handbook of child psychology* (pp. 793–828). Wiley. <https://doi.org/10.1002/9780470147658.chpsy0114>
- Burke, S. E., Sanson, A. V., & Van Hoorn, J. (2018). The psychological effects of climate change on children. *Current Psychiatry Reports*, 20(5), 1–8. <https://doi.org/10.1007/s11920-018-0896-9>
- Buthmann, J., Ham, J., Davey, K., Finik, J., Dana, K., Pehme, P., Zhang, W., Glover, V., & Nomura, Y. (2019). Infant temperament: Repercussions of superstorm Sandy-related maternal stress. *Child Psychiatry & Human Development*, 50(1), 150–162. <https://doi.org/10.1007/s10578-018-0828-2>
- Campbell, A., & Spencer, N. (2021). The macroeconomic impact of extreme weather: Evidence from Jamaica. *International Journal of Disaster Risk Reduction*, 61(July), 102336. <https://doi.org/10.1016/j.ijdrr.2021.102336>
- Center on the Developing Child at Harvard University. (2016). *Building core capabilities for life: The science behind the skills adults need to succeed in parenting and in the workplace*. www.developingchild.harvard.edu
- Cicchetti, D. (2013). Annual research review: Resilient functioning in maltreated children—Past, present, and future perspectives. *Journal of Child Psychology and Psychiatry*, 54(4), 402–422. <https://doi.org/10.1111/j.1469-7610.2012.02608.x>
- Conger, K. J., Rueter, M. A., & Conger, R. D. (2000). The role of economic pressure in the lives of parents and their adolescents: The family stress model. In L. J. Crockett & R. K. Silbereisen (Eds.), *Negotiating adolescence in times of social change* (pp. 201–223). Cambridge University Press. <https://psycnet.apa.org/record/1999-04390-012>
- Couttenier, M., & Soubeyran, R. (2013). Drought and civil war in sub-Saharan Africa. *The Economic Journal*, 124(575), 201–244. <https://doi.org/10.1111/eoj.12042>
- Cuartas, J., McCoy, D., Sánchez, J., Behrman, J., Cappa, C., Donati, G., Heymann, J., Lu, C., Raikes, A., Rao, N., Richter, L., Stein, A., & Yoshikawa, H. (2023). Family play, reading, and other stimulation and early childhood development in five low-and-middle-income

- countries. *Developmental Science*, 26(6), e13404. <https://doi.org/10.1111/desc.13404>
- Cuartas, J., McCoy, D. C., & Molano, A. (2018). The acute effect of community violent crime on maternal engagement in cognitive and socioemotional stimulation. *Early Childhood Research Quarterly*, 45, 143–154. <https://doi.org/10.1016/j.ecresq.2018.06.001>
- Cuellar, J., Jones, D. J., & Sterrett, E. (2015). Examining parenting in the neighborhood context: A review. *Journal of Child and Family Studies*, 24(1), 195–219. <https://doi.org/10.1007/s10826-013-9826-y>
- Dodge, K. A. (1993). Social-cognitive mechanisms in the development of conduct disorder and depression. *Annual Review of Psychology*, 44(1), 559–584. <https://doi.org/10.1146/annurev.ps.44.020193.003015>
- Duchoslav, J. (2017). Prenatal temperature shocks reduce cooperation: Evidence from public goods games in Uganda. *Frontiers in Behavioral Neuroscience*, 11, 249. <https://doi.org/10.3389/fnbeh.2017.00249>
- Duncan, G., Magnuson, K. A., & Votruba-Drzal, E. (2015). Children and socioeconomic status. In M. H. Bornstein, T. Leventhal, & R. M. Lerner (Eds.), *Handbook of child psychology and developmental science* (pp. 534–573). Wiley & Sons. <https://doi.org/10.1002/9781118963418.childpsy414>
- Gao, Y., Zhang, L., Kc, A., Wang, Y., Zou, S., Chen, C., Huang, Y., Mi, X., & Zhou, H. (2021). Housing environment and early childhood development in sub-Saharan Africa: A cross-sectional analysis. *PLoS Medicine*, 18(4), e1003578. <https://doi.org/10.1371/journal.pmed.1003578>
- Gartland, D., Riggs, E., Muyeen, S., Giallo, R., Afifi, T. O., MacMillan, H., Herrman, H., Bulford, E., & Brown, S. J. (2019). What factors are associated with resilient outcomes in children exposed to social adversity? A systematic review. *BMJ Open*, 9(4), e024870. <https://doi.org/10.1136/bmjopen-2018-024870>
- Gaspar, F. W., Harley, K. G., Kogut, K., Chevrier, J., Mora, A. M., Sjödin, A., & Eskenazi, B. (2015). Prenatal DDT and DDE exposure and child IQ in the CHAMACOS cohort. *Environment International*, 85(December), 206–212. <https://doi.org/10.1016/j.envint.2015.09.004>
- Ghimire, R., & Ferreira, S. (2016). Floods and armed conflict. *Environment and Development Economics*, 21(1), 23–52. <https://doi.org/10.1017/S1355770X15000157>
- Guo, C., Chen, G., He, P., Zhang, L., & Zheng, X. (2020). Risk of cognitive impairment in children after maternal exposure to the 1998 Yangtze River flood during pregnancy: Analysis of data from China's second National Sample Survey on disability. *Lancet Planetary Health*, 4(11), e522–e529. [https://doi.org/10.1016/s2542-5196\(20\)30198-4](https://doi.org/10.1016/s2542-5196(20)30198-4)
- Heilmann, A., Mehay, A., Watt, R. G., Kelly, Y., Durrant, J. E., van Turnhout, J., & Gershoff, E. T. (2021). Physical punishment and child outcomes: A narrative review of prospective studies. *The Lancet*, 398(10297), P355–P364. [https://doi.org/10.1016/S0140-6736\(21\)00582-1](https://doi.org/10.1016/S0140-6736(21)00582-1)
- Helldén, D., Andersson, C., Nilsson, M., Ebi, K. L., Friberg, P., & Alfvén, T. (2021). Climate change and child health: A scoping review and an expanded conceptual framework. *The Lancet Planetary Health*, 5(3), e164–e175. [https://doi.org/10.1016/S2542-5196\(20\)30274-6](https://doi.org/10.1016/S2542-5196(20)30274-6)
- Hensch, T. K. (2005). Critical period plasticity in local cortical circuits. *Nature Reviews Neuroscience*, 6(11), 877–888. <https://doi.org/10.1038/nrn1787>
- Hocher, B. (2014). More than genes: The advanced fetal programming hypothesis. *Journal of Reproductive Immunology*, 104–105(October), 8–11. <https://doi.org/10.1016/j.jri.2014.03.001>
- Hsiang, S. M., Burke, M., & Miguel, E. (2013). Quantifying the influence of climate on human conflict. *Science*, 341(6151), 1235367. <https://doi.org/10.1126/science.1235367>
- Ibarrarán, M. E., Ruth, M., Ahmad, S., & London, M. (2009). Climate change and natural disasters: Macroeconomic performance and distributional impacts. *Environment, Development and Sustainability*, 11(3), 549–569. <https://doi.org/10.1007/s10668-007-9129-9>
- Jensen, S. K. G., Berens, A. E., & Nelson, C. A., 3rd. (2017). Effects of poverty on interacting biological systems underlying child development. *The Lancet Child & Adolescent Health*, 1(3), 225–239. [https://doi.org/10.1016/S2352-4642\(17\)30024-X](https://doi.org/10.1016/S2352-4642(17)30024-X)
- Jeong, J., Franchett, E. E., Ramos de Oliveira, C. V., Rehmani, K., & Yousafzai, A. K. (2021). Parenting interventions to promote early child development in the first three years of life: A global systematic review and meta-analysis. *PLoS Medicine*, 18(5), e1003602. <https://doi.org/10.1371/journal.pmed.1003602>
- Jeong, J., McCoy, D. C., & Fink, G. (2017). Pathways between paternal and maternal education, caregivers' support for learning, and early child development in 44 low- and middle-income countries. *Early Childhood Research Quarterly*, 41, 136–148. <https://doi.org/10.1016/j.ecresq.2017.07.001>
- Kelley, C. P., Mohtadi, S., Cane, M. A., Seager, R., & Kushnir, Y. (2015). Climate change in the Fertile Crescent and implications of the recent Syrian drought. *Proceedings of the National Academy of Sciences of the United States of America*, 112(11), 3241–3246. <https://doi.org/10.1073/pnas.1421533112>
- Kim, S., Eom, S., Kim, H. J., Lee, J. J., Choi, G., Choi, S., Kim, S., Kim, S. Y., Cho, G., Kim, Y. D., Suh, E., Kim, S. K., Kim, S., Kim, G. H., Moon, H. B., Park, J., Kim, S., Choi, K., & Eun, S. H. (2018). Association between maternal exposure to major phthalates, heavy metals, and persistent organic pollutants, and the neurodevelopmental performances of their children at 1 to 2 years of age—Check cohort study. *Science of the Total Environment*, 624, 377–384. <https://doi.org/10.1016/j.scitotenv.2017.12.058>
- Knudsen, E. I. (2004). Sensitive periods in the development of the brain and behavior. *Journal of Cognitive Neuroscience*, 16(8), 1412–1425. <https://doi.org/10.1162/0898929042304796>
- Koepp, A. E., Lanza, K., Byrd-Williams, C., Bryan, A. E., & Gershoff, E. T. (2023). Ambient temperature increases and preschoolers' outdoor physical activity. *JAMA Pediatrics*, 177(5), 539–540. <https://doi.org/10.1001/jamapediatrics.2023.0067>
- Koubi, V. (2019). Climate change and conflict. *Annual Review of Political Science*, 22, 343–360. <https://doi.org/10.1146/annurev-polisci-050317-070830>
- Kuzik, N., Naylor, P.-J., Spence, J. C., & Carson, V. (2020). Movement behaviours and physical, cognitive, and social-emotional development in preschool-aged children: Cross-sectional associations using compositional analyses. *PLoS One*, 15(8), e0237945. <https://doi.org/10.1371/journal.pone.0237945>
- Lee, H. F., Zhang, D. D., Pei, Q., Jia, X., & Yue, R. P. H. (2017). Quantitative analysis of the impact of droughts and floods on internal wars in China over the last 500 years. *Science China Earth Sciences*, 60(12), 2078–2088. <https://doi.org/10.1007/s11430-017-9084-0>
- Leijten, P., Gardner, F., Melendez-Torres, G. J., van Aar, J., Hutchings, J., Schulz, S., Knerr, W., & Overbeek, G. (2019). Meta-analyses: Key parenting program components for disruptive child behavior. *Journal of the American Academy of Child & Adolescent Psychiatry*, 58(2), 180–190. <https://doi.org/10.1016/j.jaac.2018.07.900>
- Liao, J., Zhang, B., Xia, W., Cao, Z., Zhang, Y., Liang, S., Hu, K., Xu, S., & Li, Y. (2019). Residential exposure to green space and early childhood neurodevelopment. *Environment International*, 128(July), 70–76. <https://doi.org/10.1016/j.envint.2019.03.070>
- Lu, C., Cuartas, J., Fink, G., McCoy, D., Liu, K., Li, Z., Daelmans, B., & Richter, L. (2020). Inequalities in early childhood care and development in low/middle-income countries: 2010–2018. *BMJ Global Health*, 5(2), e002314. <https://doi.org/10.1136/bmjgh-2020-002314>
- Mach, K. J., Kraan, C. M., Adger, W. N., Buhaug, H., Burke, M., Fearon, J. D., Field, C. B., Hendrix, C. S., Maystadt, J.-F., O'Loughlin, J., Roessler, P., Scheffran, J., Schultz, K. A., & von Uexkull, N. (2019). Climate as a risk factor for armed conflict. *Nature*, 571(7764), 193–197. <https://doi.org/10.1038/s41586-019-1300-6>
- Maclean, J. C., Popovici, I., & French, M. T. (2016). Are natural disasters in early childhood associated with mental health and

- substance use disorders as an adult? *Social Science & Medicine*, 151(February), 78–91. <https://doi.org/10.1016/j.socscimed.2016.01.006>
- Masten, A. S., Motti-Stefanidi, F., & Rahl-Brigman, H. A. (2019). Developmental risk and resilience in the context of devastation and forced migration. In R. D. Parke & G. H. Elder, Jr. (Eds.), *Children in changing worlds: Sociocultural and temporal perspectives* (pp. 84–111). Cambridge University Press. <https://doi.org/10.1017/9781108264846.004>
- Maystadt, J.-F., Calderone, M., & You, L. (2014). Local warming and violent conflict in North and South Sudan. *Journal of Economic Geography*, 15(3), 649–671. <https://doi.org/10.1093/jeg/ibu033>
- McCoy, D., Cuartas, J., Behrman, J., Cappa, C., Heymann, J., López Bóo, F., Lu, C., Raikes, A., Richter, L., Stein, A., & Fink, G. (2021). Global estimates of the implications of COVID-19-related preprimary school closures for children's instructional access, development, learning, and economic wellbeing. *Child Development*, 92(5), e883–e899. <https://doi.org/10.1111/cdev.13658>
- McCoy, D. C., Peet, E. D., Ezzati, M., Danaei, G., Black, M. M., Sudfeld, C. R., Fawzi, W., & Fink, G. (2016). Early childhood developmental status in low- and middle-income countries: National, regional, and global prevalence estimates using predictive modeling. *PLoS Medicine*, 13(6), e1002034. <https://doi.org/10.1371/journal.pmed.1002034>
- McLaughlin, K. A., Sheridan, M. A., Humphreys, K. L., Belsky, J., & Ellis, B. J. (2021). The value of dimensional models of early experience: Thinking clearly about concepts and categories. *Perspectives on Psychological Science*, 16(6), 1463–1472. <https://doi.org/10.1177/1745691621992346>
- McLaughlin, K. A., Weissman, D., & Bitrán, D. (2019). Childhood adversity and neural development: A systematic review. *Annual Review of Developmental Psychology*, 1(1), 277–312. <https://doi.org/10.1146/annurev-devpsych-121318-084950>
- Miles-Novelo, A., & Anderson, C. A. (2019). Climate change and psychology: Effects of rapid global warming on violence and aggression. *Current Climate Change Reports*, 5(1), 36–46. <https://doi.org/10.1007/s40641-019-00121-2>
- Miles-Novelo, A., & Anderson, C. A. (2022). *Climate change and human behavior: Impacts of a rapidly changing climate on human aggression and violence*. Cambridge University Press. <https://doi.org/10.1017/9781108953078>
- Moss, K. M., Simcock, G., Cobham, V., Kildea, S., Elgebeili, G., Laplante, D. P., & King, S. (2017). A potential psychological mechanism linking disaster-related prenatal maternal stress with child cognitive and motor development at 16 months: The QF2011 Queensland flood study. *Developmental Psychology*, 53(4), 629–641. <https://doi.org/10.1037/dev0000272>
- Nelson, C. A., III, & Gabard-Durnam, L. J. (2020). Early adversity and critical periods: Neurodevelopmental consequences of violating the expectable environment. *Trends in Neurosciences*, 43(3), 133–143. <https://doi.org/10.1016/j.tins.2020.01.002>
- Niu, J., Liang, H., Tian, Y., Yuan, W., Xiao, H., Hu, H., Sun, X., Song, X., Wen, S., Yang, L., Ren, Y., & Miao, M. (2019). Prenatal plasma concentrations of Perfluoroalkyl and polyfluoroalkyl substances and neuropsychological development in children at four years of age. *Environmental Health*, 18(1), 53. <https://doi.org/10.1186/s12940-019-0493-3>
- Nomura, Y., Rompala, G., Pritchett, L., Aushev, V., Chen, J., & Hurd, Y. L. (2021). Natural disaster stress during pregnancy is linked to reprogramming of the placenta transcriptome in relation to anxiety and stress hormones in young offspring. *Molecular Psychiatry*, 26(11), 6520–6530. <https://doi.org/10.1038/s41380-021-01123-z>
- Obradovich, N., Migliorini, R., Paulus, M. P., & Rahwan, I. (2018). Empirical evidence of mental health risks posed by climate change. *Proceedings of the National Academy of Sciences of the United States of America*, 115(43), 10953–10958. <https://doi.org/10.1073/pnas.1801528115>
- Odo, D. B., Yang, I. A., Dey, S., Hammer, M. S., van Donkelaar, A., Martin, R. V., Dong, G. H., Yang, B. Y., Hystad, P., & Knibbs, L. D. (2023). A cross-sectional analysis of long-term exposure to ambient air pollution and cognitive development in children aged 3–4 years living in 12 low- and middle-income countries. *Environmental Pollution*, 318, 120916. <https://doi.org/10.1016/j.envpol.2022.120916>
- Palinkas, L. A., & Wong, M. (2020). Global climate change and mental health. *Current Opinion in Psychology*, 32(April), 12–16. <https://doi.org/10.1016/j.copsyc.2019.06.023>
- Patton, G. C., Sawyer, S. M., Santelli, J. S., Ross, D. A., Afifi, R., Allen, N. B., Arora, M., Azzopardi, P., Baldwin, W., Bonell, C., Kakuma, R., Kennedy, E., Mahon, J., McGovern, T., Mokdad, A. H., Patel, V., Petroni, S., Reavley, N., Taiwo, K., ... Viner, R. M. (2016). Our future: A lancet commission on adolescent health and wellbeing. *The Lancet*, 387(10036), 2423–2478. [https://doi.org/10.1016/S0140-6736\(16\)00579-1](https://doi.org/10.1016/S0140-6736(16)00579-1)
- Peterson, B. S., Bansal, R., Sawardekar, S., Nati, C., Elgalalawy, E. R., Hoepner, L. A., Garcia, W., Hao, X., Margolis, A., Perera, F., & Rauh, V. (2022). Prenatal exposure to air pollution is associated with altered brain structure, function, and metabolism in childhood. *Journal of Child Psychology and Psychiatry*, 63(11), 1316–1331. <https://doi.org/10.1111/jcpp.13578>
- Pluess, M. (2015). Individual differences in environmental sensitivity. *Child Development Perspectives*, 9(3), 138–143. <https://doi.org/10.1111/cdep.12120>
- Pörtner, H.-O., Roberts, D. C., Adams, H., Adler, C., Aldunce, P., Ali, E., Begum, R. A., Betts, R., Kerr, R. B., & Biesbroek, R. (2022). *Climate change 2022: Impacts, adaptation and vulnerability*. Intergovernmental Panel on Climate Change. https://www.ipcc.ch/report/ar6/wg2/downloads/report/IPCC_AR6_WGII_SummaryVolume.pdf
- Putra, I. G. N. E., Astell-Burt, T., Cliff, D. P., Vella, S. A., John, E. E., & Feng, X. (2020). The relationship between green space and prosocial behaviour among children and adolescents: A systematic review. *Frontiers in Psychology*, 11. <https://doi.org/10.3389/fpsyg.2020.00859>
- Rees, N. (2021). *The climate crisis is a child rights crisis: Introducing the children's climate risk index*. UNICEF.
- Romanello, M., Di Napoli, C., Drummond, P., Green, C., Kennard, H., Lampard, P., Scamman, D., Arnell, N., Ayeb-Karlsson, S., Ford, L. B., Belesova, K., Bowen, K., Cai, W., Callaghan, M., Campbell-Lendrum, D., Chambers, J., van Daalen, K. R., Dalin, C., Dasandi, N., ... Costello, A. (2022). The 2022 report of the lancet countdown on health and climate change: Health at the mercy of fossil fuels. *The Lancet*, 400(10363), 1619–1654. [https://doi.org/10.1016/S0140-6736\(22\)01540-9](https://doi.org/10.1016/S0140-6736(22)01540-9)
- Salehyan, I., & Hendrix, C. S. (2014). Climate shocks and political violence. *Global Environmental Change*, 28(September), 239–250. <https://doi.org/10.1016/j.gloenvcha.2014.07.007>
- Sameroff, A. (2010). A unified theory of development: A dialectic integration of nature and nurture. *Child Development*, 81(1), 6–22. <https://doi.org/10.1111/j.1467-8624.2009.01378.x>
- Sánchez-Ariza, J., Cuartas, J., & Moya, A. (2023). The mental health of caregivers and young children in conflict-affected settings. *AEA Papers and Proceedings*, 113(May), 336–341. <https://doi.org/10.1257/pandp.20231017>
- Sandler, I., Ingram, A., Wolchik, S., Tein, J.-Y., & Winslow, E. (2015). Long-term effects of parenting-focused preventive interventions to promote resilience of children and adolescents. *Child Development Perspectives*, 9(3), 164–171. <https://doi.org/10.1111/cdep.12126>
- Sanson, A. V., Van Hoorn, J., & Burke, S. E. L. (2019). Responding to the impacts of the climate crisis on children and youth. *Child Development Perspectives*, 13(4), 201–207. <https://doi.org/10.1111/cdep.12342>
- Sherwood, S. C., Webb, M. J., Annan, J. D., Armour, K. C., Forster, P. M., Hargreaves, J. C., Hegerl, G., Klein, S. A., Marvel, K. D., Rohling, E. J., Watanabe, M., Andrews, T., Braconnot, P.,

- Bretherton, C. S., Foster, G. L., Hausfather, Z., von der Heydt, A. S., Knutti, R., Mauritsen, T., ... Zelinka, M. D. (2020). An assessment of earth's climate sensitivity using multiple lines of evidence. *Reviews of Geophysics*, 58(4), e2019RG000678. <https://doi.org/10.1029/2019RG000678>
- Simcock, G., Cobham, V. E., Laplante, D. P., Elgbeili, G., Gruber, R., Kildea, S., & King, S. (2019). A cross-lagged panel analysis of children's sleep, attention, and mood in a prenatally stressed cohort: The QF2011 Queensland flood study. *Journal of Affective Disorders*, 255(August), 96–104. <https://doi.org/10.1016/j.jad.2019.05.041>
- Tessum, C. W., Apte, J. S., Goodkind, A. L., Muller, N. Z., Mullins, K. A., Paoletta, D. A., Polasky, S., Springer, N. P., Thakrar, S. K., Marshall, J. D., & Hill, J. D. (2019). Inequity in consumption of goods and services adds to racial-ethnic disparities in air pollution exposure. *Proceedings of the National Academy of Sciences of the United States of America*, 116(13), 6001–6006. <https://doi.org/10.1073/pnas.1818859116>
- Thelen, E., & Smith, L. B. (1998). Dynamic systems theories. In W. Damon & R. M. Lerner (Eds.), *Handbook of child psychology: Theoretical models of human development* (5th ed., pp. 563–634). John Wiley & Sons.
- UNHCR. (2022). *Global trends*. <https://www.unhcr.org/global-trends>
- United Nations. (2017). *World population prospects: The 2017 revision*. Department of Economic and Social Affairs, Population Division, United Nations.
- Valdes, V., Zorrilla, C. D., Gabard-Durnam, L., Muler-Mendez, N., Rahman, Z. I., Rivera, D., & Nelson, C. A., 3rd. (2019). Cognitive development of infants exposed to the Zika virus in Puerto Rico. *JAMA Network Open*, 2(10), e1914061. <https://doi.org/10.1001/jamanetworkopen.2019.14061>
- Vallejo, L., & Mullan, M. (2017). *Climate-resilient infrastructure: Getting the policies right*. OECD environment working papers no. 121: OECD.
- van den Dries, M. A., Ferguson, K. K., Keil, A. P., Pronk, A., Spaan, S., Ghassabian, A., Santos, S., Jaddoe, V. W. V., Trasande, L., Tiemeier, H., & Guxens, M. (2021). Prenatal exposure to nonpersistent chemical mixtures and offspring IQ and emotional and behavioral problems. *Environmental Science & Technology*, 55(24), 16502–16514. <https://doi.org/10.1021/acs.est.1c04455>
- Wachs, T. D., Black, M. M., & Engle, P. L. (2009). Maternal depression: A global threat to children's health, development, and behavior and to human rights. *Child Development Perspectives*, 3(1), 51–59. <https://doi.org/10.1111/j.1750-8606.2008.00077.x>
- Wachs, T. D., Georgieff, M., Cusick, S., & McEwen, B. S. (2014). Issues in the timing of integrated early interventions: Contributions from nutrition, neuroscience, and psychological research. *Annals of the New York Academy of Sciences*, 1308(1), 89–106. <https://doi.org/10.1111/nyas.12314>
- Walker, S., Wachs, T. D., Grantham-McGregor, S., Black, M. M., Nelson, C. A., Huffman, S. L., Baker-Henningham, H., Chang, S. M., Hamadani, J. D., Lozoff, B., Gardner, J. M. M., Powell, C. A., Rahman, A., & Richter, L. (2011). Inequality in early childhood: Risk and protective factors for early child development. *The Lancet*, 378(9799), 1325–1338. [https://doi.org/10.1016/S0140-6736\(11\)60555-2](https://doi.org/10.1016/S0140-6736(11)60555-2)
- Yoshikawa, H., & Hsueh, J. (2001). Child development and public policy: Toward a dynamic systems perspective. *Child Development*, 72(6), 1887–1903. <https://doi.org/10.1111/1467-8624.00384>
- Zare Sakhvidi, M. J., Knobel, P., Bauwelinck, M., de Keijzer, C., Boll, L. M., Spano, G., Ubalde-Lopez, M., Sanesi, G., Mehrparvar, A. H., Jacquemin, B., & Dadvand, P. (2022). Greenspace exposure and children behavior: A systematic review. *Science of the Total Environment*, 824, 153608. <https://doi.org/10.1016/j.scitotenv.2022.153608>

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Cuartas, J., McCoy, D. C., Torres, I., Burghardt, L., Shonkoff, J. P., & Yoshikawa, H. (2024). The developmental consequences of early exposure to climate change-related risks. *Child Development Perspectives*, 18, 145–154. <https://doi.org/10.1111/cdep.12503>